Discrete Mathematics and Graph Theory

Sequences and Ordering

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Sequences and Ordering

Module 3

- Sequences, Special Integer Sequences, Summations, Infinite series
- Partial Orderings, Lexicographic order, Hasse diagrams, Lattices, Topological sorting

Sequences

A sequence is a discrete structure used to represent an ordered list.
For example, 1, 2, 3, 5, 8 is a sequence with five terms and 1, 3, 9, 27, 81 , . . . , 3^n, . . . is an infinite sequence.

A *sequence* is a function from a subset of the set of integers (usually either the set {0, 1, 2, . . .} or the set {1, 2, 3, . . .}) to a set S. We use the notation a_n to denote the image of the integer n. We call a_n a *term* of the sequence.

Consider the sequence $\{a_n\}$, where

$$a_n = \frac{1}{n}.$$

The list of the terms of this sequence, beginning with a_1, namely, $a_1, a_2, a_3, a_4, \ldots,$

starts with

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots.$$

Geometric and Arithmetic Progression

A *geometric progression* is a sequence of the form

$$a, ar, ar^2, \ldots, ar^n, \ldots$$

where the *initial term* a and the *common ratio* r are real numbers.

The sequences $\{b_n\}$ with $b_n = (-1)^n$, $\{c_n\}$ with $c_n = 2 \cdot 5^n$, and $\{d_n\}$ with $d_n = 6 \cdot (1/3)^n$ are geometric progressions with initial term and common ratio equal to 1 and -1; 2 and 5; and 6 and $1/3$, respectively, if we start at $n = 0$. The list of terms $b_0, b_1, b_2, b_3, b_4, \ldots$ begins with

$$1, -1, 1, -1, 1, \ldots;$$

the list of terms $c_0, c_1, c_2, c_3, c_4, \ldots$ begins with

$$2, 10, 50, 250, 1250, \ldots;$$

and the list of terms $d_0, d_1, d_2, d_3, d_4, \ldots$ begins with

$$6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots.$$

An *arithmetic progression* is a sequence of the form

$$a, a + d, a + 2d, \ldots, a + nd, \ldots$$

where the *initial term* a and the *common difference* d are real numbers.

The sequences $\{s_n\}$ with $s_n = -1 + 4n$ and $\{t_n\}$ with $t_n = 7 - 3n$ are both arithmetic progressions with initial terms and common differences equal to -1 and 4, and 7 and -3, respectively,
if we start at n = 0. The list of terms $s_0, s_1, s_2, s_3, \ldots$ begins with
-1, 3, 7, 11, . . . ,
and the list of terms $t_0, t_1, t_2, t_3, \ldots$ begins with
7, 4, 1, -2, ...

Recurrence Relations

A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer. A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation. (A recurrence relation is said to *recursively define* a sequence.

Let {an} be a sequence that satisfies the recurrence relation an = an-1 + 3 for n = 1, 2, 3, . . . , and suppose that a0 = 2. What are a1, a2, and a3?

Solution: We see from the recurrence relation that a1 = a0 + 3 = 2 + 3 = 5. It then follows that a2 = 5 + 3 = 8 and a3 = 8 + 3 = 11. The initial conditions for a recursively defined sequence specify the terms that precede the first term where the recurrence relation takes effect. The initial condition in Example 5 is a0 = 2,

Let {an} be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for n = 2, 3, 4, . . . , and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3?

Solution: We see from the recurrence relation that $a_2 = a_1 - a_0 = 5 - 3 = 2$ and $a_3 = a_2 - a_1 = 2 - 5 = -3$. The initial conditions are $a_0 = 3$ and $a_1 = 5$.

Examples of Recurrence Relations

The *Fibonacci sequence*, $f_0, f_1, f_2, \ldots,$ is defined by the initial conditions $f_0 = 0$, $f_1 = 1$, and the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \ldots$.

Solution: The recurrence relation for the Fibonacci sequence tells us that we find successive terms by adding the previous two terms.

Because the initial conditions tell us that f0 = 0 and f1 = 1, using the recurrence relation in the definition we find that

f2 = f1 + f0 = 1 + 0 = 1, f3 = f2 + f1 = 1 + 1 = 2,

f4 = f3 + f2 = 2 + 1 = 3, f5 = f4 + f3 = 3 + 2 = 5,

f6 = f5 + f4 = 5 + 3 = 8.

Examples of Recurrence Relations

Suppose that {an} is the sequence of integers defined by $a_n = n!$, the value of the factorial function at the integer n, where n = 1, 2, 3, ...

Because $n! = n((n - 1)(n - 2) \cdot \cdot \cdot 2 \cdot 1) = n(n - 1)! = na_{n-1}$, we see that the sequence of factorials satisfies the recurrence relation $a_n = na_{n-1}$, together with the initial condition $a_1 = 1$.

Determine whether the sequence {an}, where $a_n = 3n$ for every nonnegative integer n, is a solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for n = 2, 3, 4, ... Answer the same question where an = 2n and where an = 5.

Solution: Suppose that $a_n = 3n$ for every nonnegative integer n. Then, for $n \geq 2$, we see that $2a_{n-1} - a_{n-2} = 2(3(n - 1)) - 3(n - 2) = 3n = a_n$. Therefore, {an}, where an = 3n, is a solution of the recurrence relation

Examples of Recurrence Relations

Compound Interest Suppose that a person deposits $10,000 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?

Solution: To solve this problem, let P_n denote the amount in the account after n years. Because the amount in the account after n years equals the amount in the account after n - 1 years plus interest for the nth year, we see that the sequence $\{P_n\}$ satisfies the recurrence relation

$P_n = P_{n-1} + 0.11P_{n-1} = (1.11)P_{n-1}$.

The initial condition is $P_0 = 10,000$.

We can use an iterative approach to find a formula for P_n.

Note that $P_1 = (1.11)P_0$

$P_2 = (1.11)P_1 = (1.11)_2P_0$ $P_3 = (1.11)P_2 = (1.11)_3P_0$

...

$P_n = (1.11)P_{n-1} = (1.11)_nP_0$.

When we insert the initial condition $P_0 = 10,000$, the formula $P_n = (1.11)_n10,000$ is obtained. Inserting n = 30 into the formula $P_n = (1.11)_n10,000$ shows that after 30 years the account contains

$P_{30} = (1.11)_{30}10,000 = \$228,922.97$.

Special Integer Sequences

- A common problem in discrete mathematics is finding a closed formula, a recurrence relation, or some other type of general rule for constructing the terms of a sequence.
- Sometimes only a few terms of a sequence solving a problem are known; the goal is to identify the sequence.

Find formulae for the sequences with the following first five terms: (a) 1, 1/2, 1/4, 1/8, 1/16 (b) 1, 3, 5, 7, 9 (c) 1, -1, 1, -1, 1.

Solution: (a) We recognize that the denominators are powers of 2. The sequence with $a_n = 1/2^n$, n = 0, 1, 2, . . . is a possible match. This proposed sequence is a geometric progression with a = 1 and r = 1/2.

(b) We note that each term is obtained by adding 2 to the previous term. The sequence with $a_n = 2n + 1$, n = 0, 1, 2, . . . is a possible match. This proposed sequence is an arithmetic progression with a = 1 and d = 2.

(c) The terms alternate between 1 and -1. The sequence with $a_n = (-1)^n$, n = 0, 1, 2 . . . is a possible match. This proposed sequence is a geometric progression with a = 1 and r = -1

Special Integer Sequences

1. How can we produce the terms of a sequence if the first 10 terms are 1, 2, 2, 3, 3, 3, 4, 4, 4, 4?

A reasonable rule for generating this sequence is that the integer n appears exactly n times, so the next five terms of the sequence would all be 5, the following six terms would all be 6, and so on.

2. How can we produce the terms of a sequence if the first 10 terms are 5, 11, 17, 23, 29, 35, 41, 47, 53, 59?

A reasonable guess is that the nth term is 5 + 6(n - 1) = 6n - 1. (This is an arithmetic progression with a = 5 and d = 6.)

3. How can we produce the terms of a sequence if the first 10 terms are 1, 3, 4, 7, 11, 18, 29, 47, 76, 123?

The recurrence relation is Ln = Ln-1 + Ln-2 with initial conditions L1 = 1 and L2 = 3. This sequence is known as the Lucas sequence,

Some Useful Sequences

TABLE 1 Some Useful Sequences.

nth Term	*First 10 Terms*
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, . . .
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, . . .
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, . . .
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, . . .
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, . . .
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, . . .

Conjecture a simple formula for a_n if the first 10 terms of the sequence $\{a_n\}$ are 1, 7, 25, 79, 241, 727, 2185, 6559, 19681, 59047.

We see that $a_n = 3_n - 2$. for $1 \leq n \leq 10$ and conjecture that this formula holds for all n.

Summations

We begin by describing the notation used to express the sum of the terms am, am+1, . . . , an

We use the notation $\sum_{j=m}^{n} a_j$, $\sum_{j=m}^{n} a_j$, or $\sum_{m \le j \le n} a_j$

(read as the sum from $j = m$ to $j = n$ of a_j) to represent

$$a_m + a_{m+1} + \cdots + a_n.$$

Here, the variable j is called the index of summation, and the choice of the letter j as the variable is arbitrary; that is, we could have used any other letter, such as i or k. Or, in notation

$$\sum_{j=m}^{n} a_j = \sum_{i=m}^{n} a_i = \sum_{k=m}^{n} a_k.$$

Here, the index of summation runs through all integers starting with its **lower limit** m and ending with its **upper limit** n. A large uppercase Greek letter sigma, $\sum$, is used to denote summation.

Summations

Use summation notation to express the sum of the first 100 terms of the sequence {aj }, where aj = 1/j for j = 1, 2, 3, ...

Solution: The lower limit for the index of summation is 1, and the upper limit is 100. We write this sum as

$$\sum_{j=1}^{100} \frac{1}{j}.$$

What is the value of $\sum_{j=1}^{5} j^2$?

Solution: We have

$$\begin{aligned}\sum_{j=1}^{5} j^2 &= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 \\ &= 1 + 4 + 9 + 16 + 25 \\ &= 55.\end{aligned}$$

Summations

What is the value of $\sum_{k=4}^{8}(-1)^k$?

Solution: We have

$$\sum_{k=4}^{8}(-1)^k = (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8$$

$$= 1 + (-1) + 1 + (-1) + 1$$

$$= 1.$$

Geometric Series Summation

If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^{n} ar^j = \begin{cases} \dfrac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1. \end{cases}$$

To compute S, first multiply both sides of the equality by r and then manipulate the resulting sum as follows:

$$\begin{aligned}
rS_n &= r\sum_{j=0}^{n} ar^j && \text{substituting summation formula for } S \\
&= \sum_{j=0}^{n} ar^{j+1} && \text{by the distributive property} \\
&= \sum_{k=1}^{n+1} ar^k && \text{shifting the index of summation, with } k = j+1 \\
&= \left(\sum_{k=0}^{n} ar^k\right) + (ar^{n+1} - a) && \text{removing } k = n+1 \text{ term and adding } k = 0 \text{ term} \\
&= S_n + (ar^{n+1} - a) && \text{substituting } S \text{ for summation formula}
\end{aligned}$$

Double Summations

To evaluate the double sum, first expand the inner summation and then continue by computing the outer summation:

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij = \sum_{i=1}^{4}(i + 2i + 3i)$$

$$= \sum_{i=1}^{4} 6i$$

$$= 6 + 12 + 18 + 24 = 60.$$

Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \lvert x \rvert < 1$	$\frac{1}{(1-x)^2}$

Find $\sum_{k=50}^{100} k^2$.

Solution: First note that because $\sum_{k=1}^{100} k^2 = \sum_{k=1}^{49} k^2 + \sum_{k=50}^{100} k^2$, we have

$$\sum_{k=50}^{100} k^2 = \sum_{k=1}^{100} k^2 - \sum_{k=1}^{49} k^2.$$

Using the formula $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$ from Table 2 we see that

$$\sum_{k=50}^{100} k^2 = \frac{100 \cdot 101 \cdot 201}{6} - \frac{49 \cdot 50 \cdot 99}{6} = 338{,}350 - 40{,}425 = 297{,}925.$$

Infinite Series

Let x be a real number with $|x| < 1$. Find $\sum_{n=0}^{\infty} x^n$.

Solution: By Theorem 1 with $a = 1$ and $r = x$ we see that $\sum_{n=0}^{k} x^n = \dfrac{x^{k+1} - 1}{x - 1}$. Because $|x| < 1$, x^{k+1} approaches 0 as k approaches infinity. It follows that

$$\sum_{n=0}^{\infty} x^n = \lim_{k \to \infty} \frac{x^{k+1} - 1}{x - 1} = \frac{0 - 1}{x - 1} = \frac{1}{1 - x}.$$

Partial Orderings

A relation R on a set S is called a *partial ordering* or *partial order* if it is reflexive, antisymmetric, and transitive. A set S together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R). Members of S are called *elements* of the poset.

Show that the "greater than or equal" relation (≥) is a partial ordering on the set of integers.

Solution: Because a ≥ a for every integer a, ≥ is reflexive. If a ≥ b and b ≥ a, then a = b. Hence, ≥ is antisymmetric. Finally, ≥ is transitive because a ≥ b and b ≥ c imply that a ≥ c.
It follows that ≥ is a partial ordering on the set of integers and (Z, ≥) is a poset

Totally Ordered Set

If $(S, \preccurlyeq)$ is a poset and every two elements of S are comparable, S is called a *totally ordered* or *linearly ordered set*, and $\preccurlyeq$ is called a *total order* or a *linear order*. A totally ordered set is also called a *chain*.

In the poset $(\mathbf{Z}^+, |)$, are the integers 3 and 9 comparable? Are 5 and 7 comparable?

Solution: The integers 3 and 9 are comparable, because $3 \mid 9$. The integers 5 and 7 are incomparable, because $5 \nmid 7$ and $7 \nmid 5$. ◀

The adjective "partial" is used to describe partial orderings because pairs of elements may be incomparable. When every two elements in the set are comparable, the relation is called a **total ordering**.

The poset (Z+, |) is not totally ordered because it contains elements that are incomparable, such as 5 and 7

Lexicographic Order

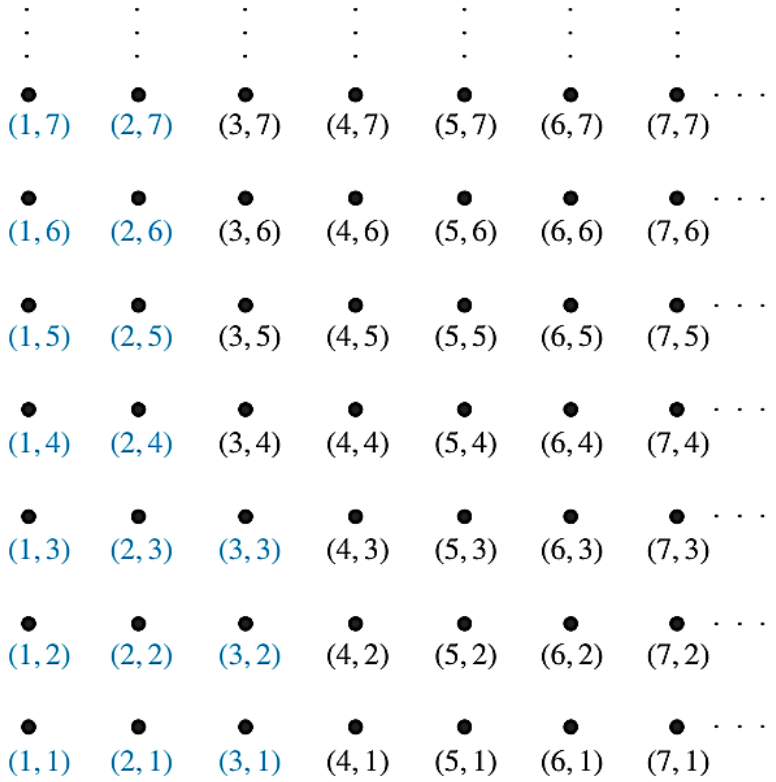
to construct a partial ordering on the Cartesian product of two posets, $(A_1, \preccurlyeq_1)$ and $(A_2, \preccurlyeq_2)$. The **lexicographic ordering** $\preccurlyeq$ on $A_1 \times A_2$ is defined by specifying that one pair is less than a second pair if the first entry of the first pair is less than (in A_1) the first entry of the second pair, or if the first entries are equal, but the second entry of this pair is less than (in A_2) the second entry of the second pair. In other words, (a_1, a_2) is less than (b_1, b_2), that is,

Determine whether $(3, 5) \prec (4, 8)$, whether $(3, 8) \prec (4, 5)$, and whether $(4, 9) \prec (4, 11)$ in the poset $(Z \times Z,)$, where is the lexicographic ordering constructed from the usual $\leq$ relation on Z.

Solution: Because $3 < 4$, it follows that $(3, 5) \prec (4, 8)$ and that $(3, 8) \prec (4, 5)$. We have $(4, 9) \prec (4, 11)$, because the first entries of (4, 9) and (4, 11) are the same but $9 < 11$.

Lexicographic Order Examples

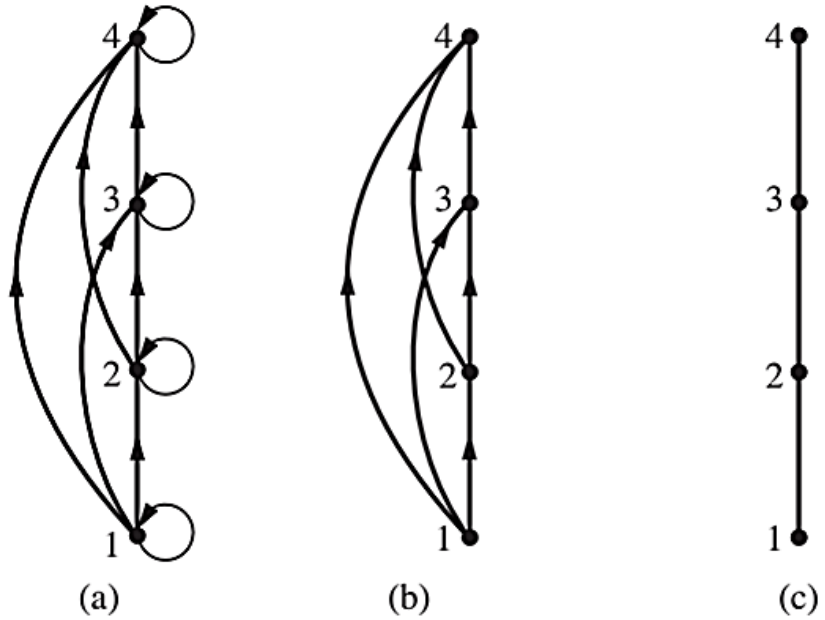
In below figure the ordered pairs in Z+× Z+ that are less than (3, 4) are highlighted



- Consider the set of strings of lowercase English letters.
- Using the ordering of letters in the alphabet, a lexicographic ordering on the set of strings can be constructed.
- A string is less than a second string if the letter in the first string in the first position where the strings differ comes before the letter in the second string in this position, or if the first string and the second string agree in all positions, but the second string has more letters.
- This ordering is the same as that used in dictionaries. For example,
- discreet ≺ discrete, because these strings differ first in the seventh position, and e ≺ t.
- Also, discreet ≺ discreetness because the first eight letters agree, but the second string is longer.
- Furthermore, discrete ≺ discretion, because discrete ≺ discreti.

Hasse Diagrams

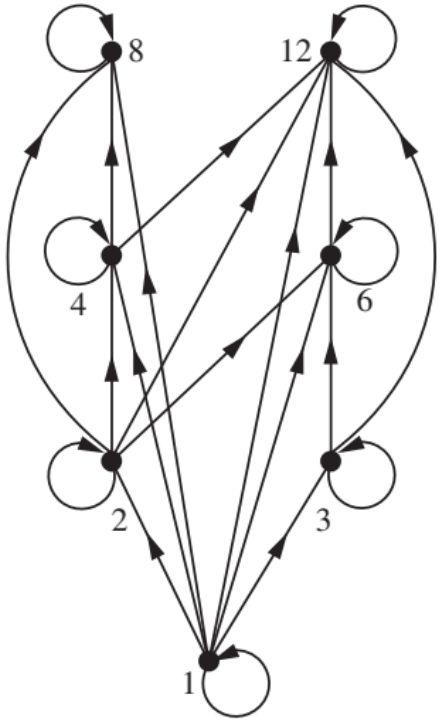
- Start with the directed graph for this relation.
- Because a partial ordering is reflexive, a loop (a, a) is present at every vertex a.
- Remove these loops.
- Next, remove all edges that must be in the partial ordering because of the presence of other edges and transitivity.
- That is, remove all edges (x, y) for which there is an element $z \in S$ such that $x \prec z$ and $z \prec x$.
- Finally, arrange each edge so that its initial vertex is below its terminal vertex (as it is drawn on paper).
- Remove all the arrows on the directed edges, because all edges point "upward" toward their terminal vertex.



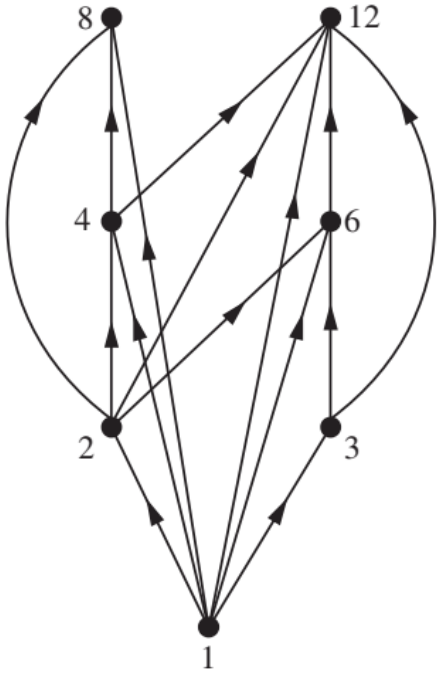
Constructing the Hasse Diagram for $(\{1, 2, 3, 4\}, \leq)$

Hasse diagrams

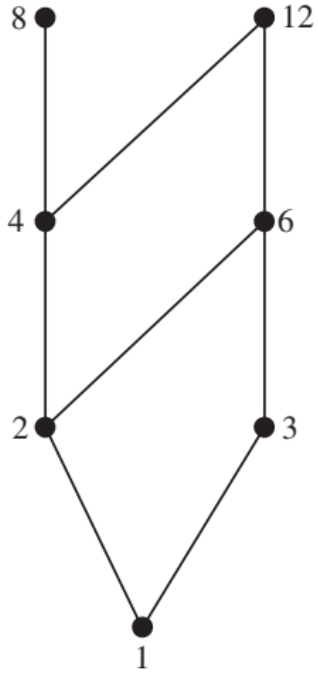
Draw the Hasse diagram representing the partial ordering {(a, b)|a divides b} on {1, 2, 3, 4, 6, 8, 12}.



(a)



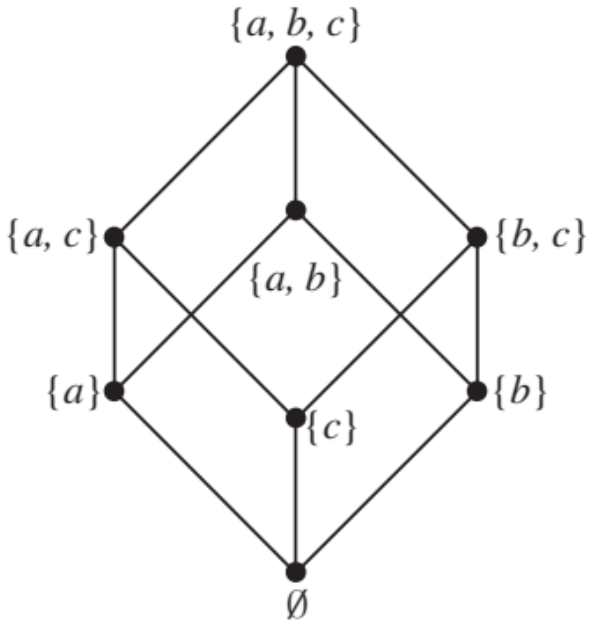
(b)



(c)

Sample Slide

Draw the Hasse diagram for the partial ordering {(A, B) | A ⊆ B} on the power set P (S) where S = {a, b, c}.

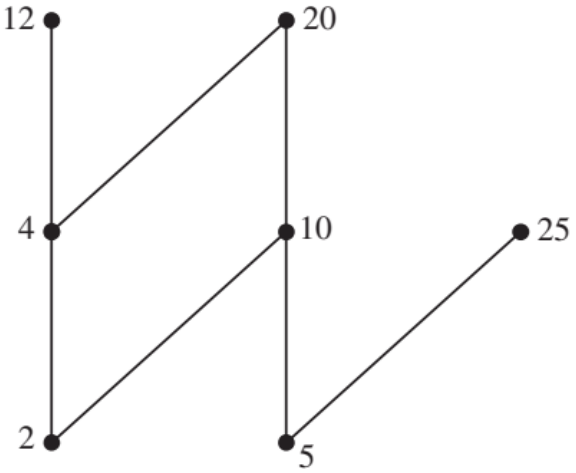


Maximal and Minimal Elements

Elements of posets that have certain extremal properties are important for many applications. An element of a poset is called maximal if it is not less than any element of the poset. That is, a is **maximal** in the poset $(S, \preccurlyeq)$ if there is no $b \in S$ such that $a \prec b$. Similarly, an element of a poset is called minimal if it is not greater than any element of the poset. That is, a is **minimal** if there is no element $b \in S$ such that $b \prec a$. Maximal and minimal elements are easy to spot using a Hasse diagram. They are the "top" and "bottom" elements in the diagram.

Which elements of the poset ({2, 4, 5, 10, 12, 20, 25}, |) are maximal, and which are minimal?

Solution: The Hasse diagram in Figure 5 for this poset shows that the maximal elements are 12, 20, and 25, and the minimal elements are 2 and 5. As this example shows, a poset can have more than one maximal element and more than one minimal element.

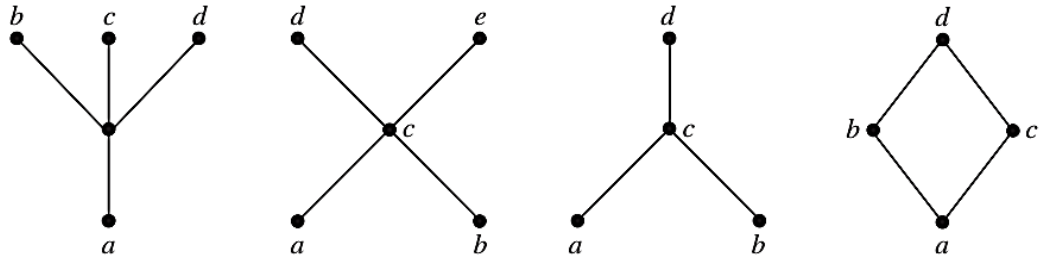


Greatest Element and Least Element

Sometimes there is an element in a poset that is greater than every other element. Such an element is called the greatest element. That is, a is the **greatest element** of the poset $(S, \preccurlyeq)$ if $b \preccurlyeq a$ for all $b \in S$. The greatest element is unique when it exists [see Exercise 40(a)]. Likewise, an element is called the least element if it is less than all the other elements in the poset. That is, a is the **least element** of $(S, \preccurlyeq)$ if $a \preccurlyeq b$ for all $b \in S$. The least element is unique

Determine whether the posets represented by each of the Hasse diagrams in Figure have a greatest element and a least element.

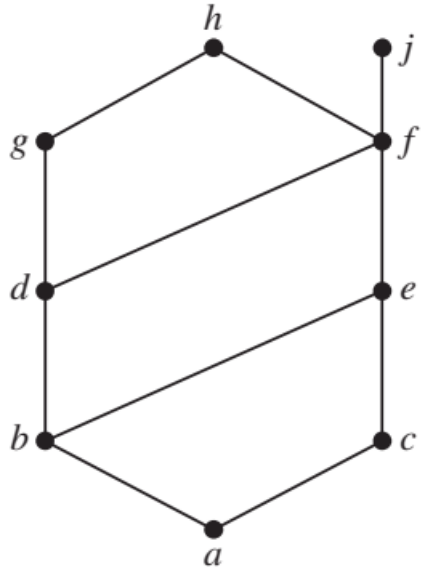
Solution: The least element of the poset with Hasse diagram (a) is a. This poset has no greatest element. The poset with Hasse diagram (b) has neither a least nor a greatest element. The poset with Hasse diagram (c) has no least element. Its greatest element is d. The poset with Hasse diagram (d) has least element a and greatest element d.



Upper Bound and Lower Bound

Sometimes it is possible to find an element that is greater than or equal to all the elements in a subset A of a poset $(S, \preccurlyeq)$. If u is an element of S such that $a \preccurlyeq u$ for all elements $a \in A$, then u is called an **upper bound** of A. Likewise, there may be an element less than or equal to all the elements in A. If l is an element of S such that $l \preccurlyeq a$ for all elements $a \in A$, then l is called a **lower bound** of A.

Find the lower and upper bounds of the subsets {a, b, c}, {j, h}, and {a, c, d, f } in the poset with the Hasse diagram shown in Figure



The upper bounds of {a, b, c} are e, f, j, and h, and its only lower bound is a. Thereare no upper bounds of {j, h}, and its lower bounds are a, b, c, d, e, and f . The upper bounds of {a, c, d, f } are f , h, and j, and its lower bound is a.

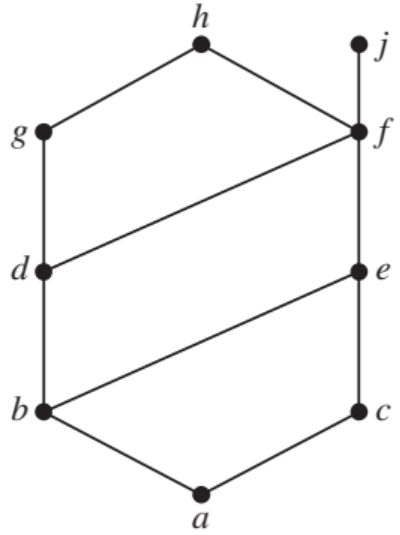
Least upper bound and Greatest lower bound

The element x is called the **least upper bound** of the subset A if x is an upper bound that is less than every other upper bound of A. Because there is only one such element, if it exists,

Similarly, the element y is called the **greatest lower bound** of A if y is a lower bound of A and $z \preccurlyeq y$ whenever z is a lower bound of A. The greatest lower bound of A is unique if it exists

Find the greatest lower bound and the least upper bound of {b, d, g}, if they exist, in the poset shown in Figure 7.

Solution: The upper bounds of {b, d, g} are g and h. Because g $\prec$ h, g is the least upper bound. The lower bounds of {b, d, g} are a and b. Because a $\prec$ b, b is the greatest lower bound.



Least upper bound and Greatest lower bound Examples

Find the greatest lower bound and the least upper bound of the sets {3, 9, 12} and {1, 2, 4, 5, 10}, if they exist, in the poset $(Z_+, |)$.

Solution: An integer is a lower bound of {3, 9, 12} if 3, 9, and 12 are divisible by this integer.

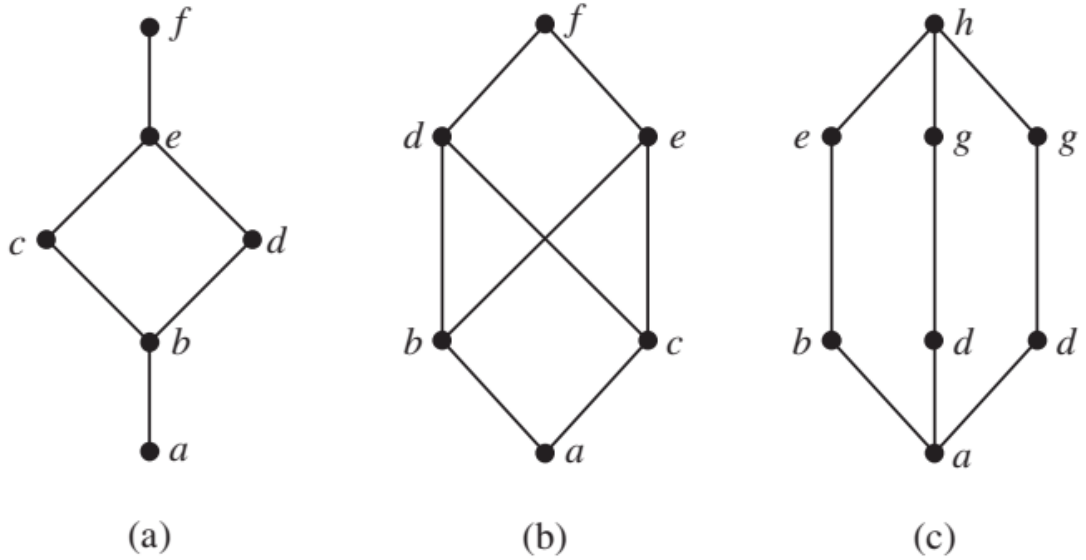
- The only such integers are 1 and 3. Because 1 | 3, 3 is the greatest lower bound of {3, 9, 12}.
- The only lower bound for the set {1, 2, 4, 5, 10} with respect to | is the element 1. Hence, 1 is the greatest lower bound for {1, 2, 4, 5, 10}.
- An integer is an upper bound for {3, 9, 12} if and only if it is divisible by 3, 9, and 12.
- The integers with this property are those divisible by the least common multiple of 3, 9, and 12, which is 36.
- Hence, 36 is the least upper bound of {3, 9, 12}. A positive integer is an upper bound for the set {1, 2, 4, 5, 10} if and only if it is divisible by 1, 2, 4, 5, and 10.
- The integers with this property are those integers divisible by the least common multiple of these integers, which is 20. Hence, 20 is the least upper bound of {1, 2, 4, 5, 10}.

Lattices

A partially ordered set in which every pair of elements has both a least upper bound and a greatest lower bound is called a lattice.

Determine whether the posets represented by each of the Hasse diagrams in Figure are lattices.

Solution: The posets represented by the Hasse diagrams in (a) and (c) are both lattices because in each poset every pair of elements has both a least upper bound and a greatest lower bound,.
The poset with the Hasse diagram shown in (b) is not a lattice, because the elements b and c have no least upper bound. Note that each of the elements d, e, and f is an upper bound, but none of these three elements precedes the other two with respect to the ordering of this poset.



Lattices Examples

Determine whether the posets ({1, 2, 3, 4, 5}, |) and ({1, 2, 4, 8, 16}, |) are lattices.
Solution: Because 2 and 3 have no upper bounds in ({1, 2, 3, 4, 5}, |), they certainly do not have a least upper bound. Hence, the first poset is not a lattice.
Every two elements of the second poset have both a least upper bound and a greatest lower bound.
The least upper bound of two elements in this poset is the larger of the elements and the greatest lower bound of two elements is the smaller of the elements. Hence, this second poset is a lattice.

Determine whether (P (S), ⊆) is a lattice where S is a set.
Solution: Let A and B be two subsets of S. The least upper bound and the greatest lower bound of A and B are A ∪ B and A ∩ B, respectively, as the reader can show. Hence, (P (S), ⊆) is a lattice.

Topological Sorting

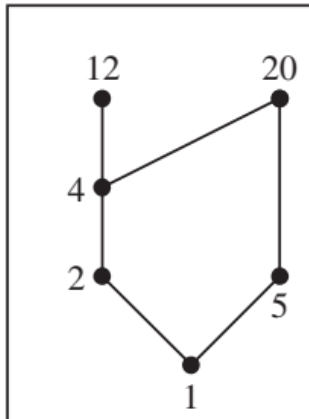
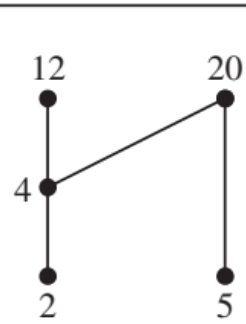
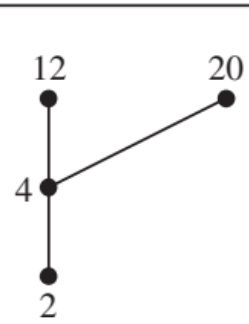
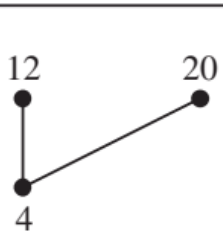
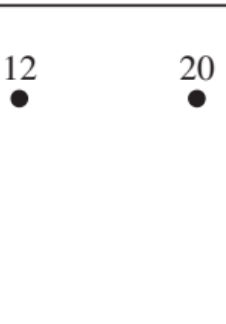
ALGORITHM 1 Topological Sorting.

procedure *topological sort* ($(S, \preceq)$: finite poset)
$k := 1$
while $S \neq \emptyset$
    $a_k :=$ a minimal element of S {such an element exists by Lemma 1}
    $S := S - \{a_k\}$
    $k := k + 1$
return $a_1, a_2, \ldots, a_n$ {$a_1, a_2, \ldots, a_n$ is a compatible total ordering of S}

Topological Sorting Examples

Find a compatible total ordering for the poset ({1, 2, 4, 5, 12, 20}, |).

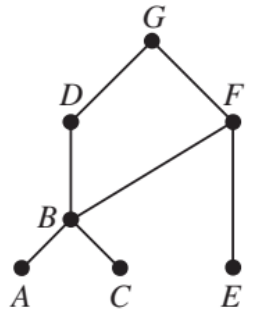
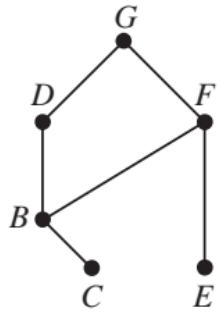
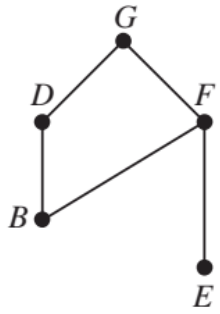
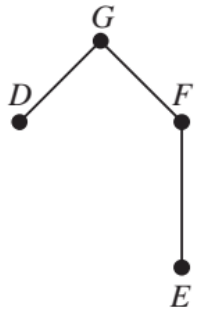
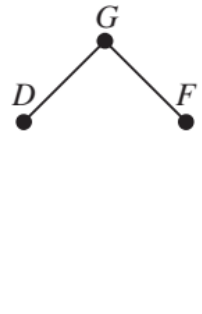
Solution: The first step is to choose a minimal element. This must be 1, because it is the only minimal element. Next, select a minimal element of ({2, 4, 5, 12, 20}, |). There are two minimal elements in this poset, namely, 2 and 5. We select 5. The remaining elements are {2, 4, 12, 20}. The only minimal element at this stage is 2. Next, 4 is chosen because it is the only minimal element of ({4, 12, 20}, |). Because both 12 and 20 are minimal elements of ({12, 20}, |), either can be chosen next. We select 20, which leaves 12 as the last element left. This produces the total ordering $1 \prec 5 \prec 2 \prec 4 \prec 20 \prec 12$

					
Minimal element chosen: 1	5	2	4	20	12

Topological Sorting Examples

A development project at a computer company requires the completion of seven tasks. Some of these tasks can be started only after other tasks are finished. A partial ordering on tasks is set up by considering task X ≺ task Y if task Y cannot be started until task X has been completed. The Hasse diagram for the seven tasks, with respect to this partial ordering, is shown in Figure 10. Find an order in which these tasks can be carried out to complete the project.

Solution: An ordering of the seven tasks can be obtained by performing a topological sort. The steps of a sort are illustrated in Figure 11. The result of this sort, A ≺ C ≺ B ≺ E ≺ F ≺ D ≺ G, gives one possible order for the tasks.

						
Minimal element chosen A	C	B	E	F	D	G

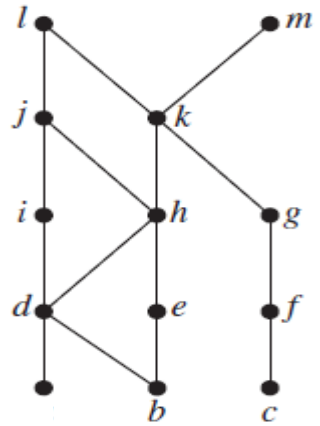
Tutorial 5

1. Which of these relations on {0, 1, 2, 3} are partial orderings? Determine the properties of a partial ordering that the others lack.

a) {(0, 0), (1, 1), (2, 2), (3, 3)} b) {(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)}

2. Answer these questions for the partial order represented by this Hasse diagram.

a) Find the maximal elements.
b) Find the minimal elements.
c) Is there a greatest element?
d) Is there a least element? e) Find all upper bounds of {a, b, c}.
f) Find the least upper bound of {a, b, c}, if it exists.
g) Find all lower bounds of {f, g, h}.
h) Find the greatest lower bound of {f, g, h}, if it exists.



3. Draw the Hasse diagram for divisibility on the set

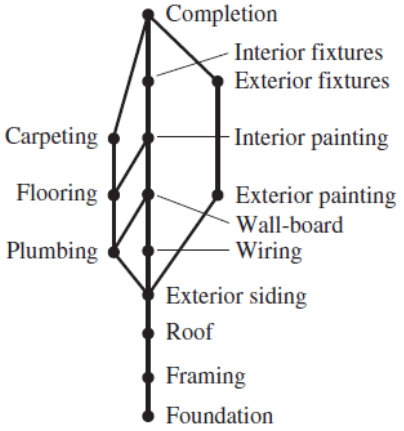
a) {1, 2, 3, 4, 5, 6}. b) {3, 5, 7, 11, 13, 16, 17}. c) {2, 3, 5, 10, 11, 15, 25}.

b) d) {1, 3, 9, 27, 81, 243}.

4. Find the lexicographic ordering of these strings of lowercase English letters:

a) quack, quick, quicksilver, quicksand, quacking

b) b) open, opener, opera, operand, opened c) zoo, zero, zoom, zoology, zoological



5. Schedule the tasks needed to build a house, by specifying their order, if the Hasse diagram representing these tasks is as shown in the figure.

Tutorial 6

1. Find these terms of the sequence $\{a_n\}$where $a_n = 2.(-3)^n + 5^n$. (i) a_0 (ii) a_1 (iii) a_4 (iv) a_5

2. List the first 10 terms of each of these sequences. **a)** the sequence that begins with 2 and in which each successive term is 3 more than the preceding term **b)** the sequence that lists each positive integer three times, in increasing order

3. For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence. **a)** 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, . . . **b)** 1, 2, 2, 3, 4, 4, 5, 6, 6, 7, 8, 8, . . .

4. What are the values of these sums? (i) $\sum_{k=1}^{5}(k+1)$ (ii)$\sum_{j=0}^{4}(-2)^j$

5. Suppose that the number of bacteria in a colony triples every hour.
a) Set up a recurrence relation for the number of bacteria after *n* hours have elapsed.
b) If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?

THANK YOU